WISEConv: A Worklist-Driven Masked-Convolution Runtime for GPUs

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Abstract

Masked convolution promises to cut the per-frame inference latency of vision models by computing only positions an upstream policy marks active, yet existing paths sacrifice either throughput or useful-compute ratio and fail to convert sparsity into proportional latency reduction. We present WISEConv, a masked-convolution runtime that replaces the dense tiled pipeline's contiguous coordinate stream with an active-output worklist derived from the upstream mask; it restricts computation to active positions while retaining dense tensor layouts and the regular per-position reduction. Securing both a high useful-compute ratio and high throughput requires amortizing construction overhead and sustaining compute throughput as per-frame worklist lengths vary. WISEConv therefore co-designs construction and consumption: compatible operators reuse the constructed worklist without reconstruction, the worklist emission order preserves tile-local spatial reuse, and a data-driven adaptive decomposition adjusts parallel work to the activity level without host synchronization. Across four perception models on six GPU platforms (four discrete, two embedded), WISEConv reduces full-model latency by 22.1–46.5% against the fastest competing path and per-frame energy by 28.5–72.6% relative to dense execution on the embedded GPUs.

Keywords: Sparse convolution, dynamic spatial sparsity, GPU runtime, deep learning systems, efficient inference

1 Introduction

Dense convolution dominates the per-frame inference latency of many latency-critical vision tasks [2, 9, 38], yet on a given input much of that computation is spatially redundant. A common remedy is masked convolution: an upstream policy marks active positions, and the operator computes only those. The mask may derive from event-camera increments [7, 23], temporal differences [16, 28], or learned spatial gates [21, 35]. Regardless of source, it poses one operator-level problem: compute the convolution only at the active positions of the dense feature grid.

Dense GPU convolution obtains high throughput from tiled, coordinate-driven execution. Each tile enumerates a contiguous block of output coordinates; those coordinates fix each output position's receptive field and write location, while the reduction over channels and kernel offsets follows a regular datapath. Neighboring coordinates share overlapping receptive fields, yielding predictable memory access and data reuse. However, unlike static sparsity in matrix multiplication, masked convolution selects positions at runtime per frame, requiring the operator to access scattered, irregularly-overlapping neighborhoods and handle a varying workload, conflicting with tiled execution.

Existing masked-convolution systems preserve this regularity or eliminate wasted work, but not both. *Tile skipping* [7, 16, 28, 30] retains the dense pipeline and suppresses a tile only when all its outputs are inactive, so a single active position triggers computation for the entire tile. *Gather-scatter* [21, 35, 39] (as an execution path) instead extracts active positions exactly, materializes receptive fields, and scatters the computation results to the dense grid, sacrificing regularity and incurring extraction, irregular access, and writeback overhead.

We characterize this trade-off along two axes: *throughput* (operations per second) and *useful-compute ratio* (the fraction of issued operations that produce needed outputs). In one of our workloads (FireFlowNet [26] under an event camera [7] mask policy, §2.2), the masks render 73.7% of the convolution work unnecessary, yet neither approach turns this into proportional latency reduction. Tile skipping [28] still wastes 26.4% of its issued operations and only cuts latency by at most 42.0%. Gather-scatter drives the useful-compute ratio near one but sustains only 21.7% of tile skipping's throughput, running 1.97× slower than the dense path.

Our key insight is that position-level selectivity does not require abandoning the dense convolution pipeline. A dense tile already derives its reads and writes from output coordinates, so replacing its contiguous coordinate block with an active-output worklist changes which outputs are computed without changing the per-position reduction, the dense tensor layout, or the weight layout. With this substitution, the operator runs as a dense tiled pipeline, raising the useful-compute ratio toward one without materializing input patches.

Translating this into latency reduction requires handling two challenges. First, *worklist construction cost*: construction scans the mask, so its cost scales with the layer's spatial grid, not with the active count or channel widths, while the compute it enables shrinks with both. Construction's share of latency therefore grows as activity falls and channels thin, and accumulates across operators when every layer rebuilds. Second, *end-to-end worklist efficiency*: the worklist's order and length both affect its consumption throughput. An unordered worklist scatters the input neighborhoods that adjacent work touches, lowering locality; its length falls below the dense grid and shifts with layer and input, so no fixed tiling keeps the GPU full, and because the length is device-resident, any host-side decision that depends on it

costs a synchronization on the per-frame path. Both effects sharpen as activity falls.

We present WISEConv (**W**orklist-dr**I**ven ma**S**k**E**d **Conv**olution), a masked-convolution runtime that fuses position-level selectivity into the dense tiled pipeline, securing both a high useful-compute ratio and high throughput by co-designing worklist construction and consumption. Construction is cheap and infrequent: a single mask-space pass fuses output-mask propagation with per-tile compaction, operating on only mask and coordinate metadata; compatible operators reuse valid metadata instead of reconstructing it. Consumption is dense-like: construction emits contiguous per-tile segments ordered by the dense tile traversal, giving the compute stage tile-local structure without global sorting. Compute then decomposes parallel work adaptively according to device, layer shape, and activity, choosing each operator's configuration from offline calibration. For temporally correlated inputs, it tracks activity across frames and sustains GPU utilization without host synchronization.

This paper makes the following contributions:

- We characterize masked-convolution latency along two axes, useful-compute ratio and throughput, exposing why existing work does not translate upstream sparsity into proportional latency reduction.
- We design WISEConv, which fuses position-level selectivity into the dense tiled convolution pipeline and co-designs worklist construction and consumption to secure both axes jointly.
- We evaluate WISEConv on four perception models across four discrete GPUs and two Jetson platforms. WISEConv reduces full-model latency relative to the fastest baseline by 34.2–46.5% on the discrete GPUs and 22.1–38.6% on the Jetson platforms, and cuts per-frame energy over dense convolution by 28.5–72.6% on the Jetson platforms.

2 Background and Motivation

2.1 Masked Convolution

Convolution runs inside the perception-action loop of many vision applications [2, 32, 36, 38, 42], where per-frame inference latency bounds system responsiveness [8–10, 17]. To cut this latency, masked convolution exploits dynamic spatial sparsity: an upstream policy emits a spatial mask, and the operator computes only the marked positions. The mask comes from sources including event-camera increments [7, 23], temporal residuals between video frames [3, 16, 27, 28, 37], or learned input-dependent gates [12, 21, 35, 39]. The event and temporal policies mark positions that changed since the previous frame; the learned gates mark positions the task treats as relevant. These sources differ in sensing modality and policy, but all expose the same dynamic spatial sparsity on a dense 2D feature grid, so a single masked-convolution runtime serves them all.

Masked convolution keeps the tensors and arithmetic of dense convolution and restricts only which output coordinates it computes. It operates on an input feature tensor $\mathbf{x} \in \mathbb{R}^{H_{in} \times W_{in} \times C_{in}}$, a weight tensor $\mathbf{w} \in \mathbb{R}^{k \times k \times C_{in} \times C_{out}}$ of kernel size k, and an output $\mathbf{y} \in \mathbb{R}^{H_{out} \times W_{out} \times C_{out}}$ on a coordinate grid $\mathcal{G} = \{1, \ldots, H_{out} W_{out}\}$, where each coordinate $p \in \mathcal{G}$ indexes a spatial position (h, w) and has a $k \times k$ receptive field $\mathcal{N}(p)$ in $\mathbf{x}$. An upstream policy marks the coordinates to compute in an output mask $M_{out} \in \{0,1\}^{H_{out} \times W_{out}}$. Spatial gating predicts M_{out} directly; event and temporal policies instead supply an input mask $M_{in} \in \{0,1\}^{H_{in} \times W_{in}}$ whose active positions propagate their influence along the receptive field, so p stays active whenever any input in $\mathcal{N}(p)$ is active,

$$M_{out}[p] = \begin{cases} 1, & \exists\, r \in \mathcal{N}(p) \text{ s.t. } M_{in}[r] = 1, \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

M_{out} thus identifies the output positions that the execution path is required to compute, each by aggregating $\mathbf{x}$ over $\mathcal{N}(p)$ with the weights $\mathbf{w}$.

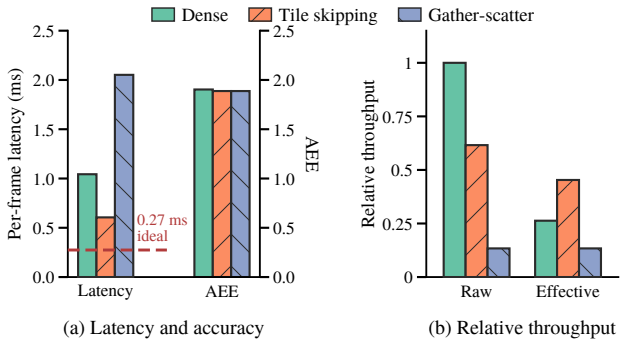
An execution path issues computation over a sequence of position slots C, where each slot is drawn from $\mathcal{G} \cup \{\perp\}$; $\perp$ denotes a padding slot that carries no output coordinate. The path must satisfy the *coverage contract*: every active coordinate is included, $\{p : M_{out}[p] = 1\} \subseteq \{c \in C : c \neq \perp\}$. For each non-$\perp$ slot $p \in C$ it computes

$$\mathbf{y}[p] = \sum_{i=1}^{k^2} \mathbf{x}[\mathcal{N}_i(p)]\, \mathbf{w}_i, \tag{2}$$

where $\mathcal{N}_i(p)$ is the i-th position in $\mathcal{N}(p)$ and $\mathbf{w}_i \in \mathbb{R}^{C_{in} \times C_{out}}$ the corresponding weight slice. How unrequested positions are represented or merged into a dense result is policy-specific. Event and temporal execution retains prior state, whereas learned gating merges the computed branch values into a residual or bypass tensor. Coverage is therefore the operator-level correctness condition: every position requested by the policy is computed, while the policy determines the treatment of all other positions. Beyond coverage, a path may issue inactive or $\perp$ slots; dense convolution is the extreme where $\{c \in C : c \neq \perp\} = \mathcal{G}$. The policy fixes the accuracy and required work, whereas the execution path determines the issued sequence C.

2.2 The Sparsity-to-Latency Gap

Existing systems execute a mask along one of two paths. *Tile skipping* [7, 16, 22, 27, 30, 37] inherits the dense pipeline's tile-based computation, skipping a tile only when all its coordinates are inactive and otherwise computing the whole tile; DeltaCNN [28] fuses a selective path into the kernel, taken when a tile is estimated to be mostly empty, to cut wasted work. *Gather-scatter* [21, 39] instead extracts exactly the active coordinates, computes, and scatters them back, removing the inactive arithmetic but adding extraction, irregular access, and writeback overhead; DynConv [35] adopts



(a) Latency and accuracy (b) Relative throughput

Figure 1. FireFlowNet [26] using the Ev-Conv [7] event-driven mask policy on MVSEC [43] and RTX 3080, comparing cuDNN (Dense), DeltaCNN tile skipping [28], and an MMCV-derived gather-scatter path [4]. (a) Mean full-model per-frame latency and average endpoint error (AEE; lower is better). The dashed line marks dense latency scaled by the active ratio aggregated across layers; the sparse paths use the same masks and produce nearly identical AEE. (b) Raw throughput T and effective throughput T_{eff}, each normalized to dense raw throughput. The dense useful-compute ratio equals the active ratio set by the Ev-Conv mask policy ($\alpha \approx 26.3\%$).

a model structure suited to gather-scatter, where one gather feeds several consecutive layers before a single scatter, amortizing that overhead.

These paths fail for different reasons, so we separate policy sparsity from execution efficiency. The policy fixes its active ratio α, the fraction of output-grid positions marked active. An execution path instead determines its useful-compute ratio η and throughput T, which combine into effective throughput T_{eff}:

$$\alpha = \frac{\|M_{out}\|_0}{|\mathcal{G}|}, \qquad \eta = \frac{\|M_{out}\|_0}{|C|} \le 1, \qquad T = \frac{2|C|k^2 C_{in} C_{out}}{t}, \quad T_{\text{eff}} = \eta T. \tag{3}$$

For $|C| > 0$, η captures inactive and padding slots in the issued schedule, while T captures how quickly the path processes those slots. The latency t includes all auxiliary mask and coordinate work, so such work lowers T without changing η; the factor of two counts each multiply-accumulate as two FLOPs. Consequently, T_{eff} measures the rate of required-output computation and therefore determines its latency.

Figure 1 traces this on RTX 3080 for FireFlowNet [26] under the Ev-Conv [7] mask policy. Event cameras expose microsecond-scale changes, and Ev-Conv evaluates optical flow on event-window inputs updated at 1 kHz. The policy marks 73.7% of the convolution work unnecessary, and the sparse paths discard it without reducing accuracy. Scaling dense latency by the fraction of convolution work that remains gives a 0.274 ms ideal proportional-scaling reference (Fig. 1a, dashed). Neither path approaches it: tile skipping (DeltaCNN [28]) reaches 0.606 ms, still 2.21× the reference, and gather-scatter (MMCV [4]) runs at 2.052 ms, almost twice the latency of dense convolution itself. The gap traces to low effective throughput on both paths. Tile skipping holds throughput at 61.6% of dense but at a useful-compute ratio of only 73.6%; gather-scatter lifts the useful-compute ratio to one, yet extraction and data movement collapse its throughput to 13.4% of dense (Fig. 1b). For masked convolution, neither path efficiently converts the GPU's capability into effective throughput.

2.3 Distinction from Related Sparse Execution

Related sparse-execution work differs from WISEConv's masked convolution in representation, operator, sparsity type, or platform. 3D sparse convolution (SpConv v2 [31], TorchSparse++ [33, 34], Minuet [40], MinkowskiEngine [6], submanifold sparse convolution [15]) is built to organize computation over an unordered list of active voxels, constructing coordinate maps that route irregular input coordinates to outputs; its input is sparse by construction rather than a dense grid under a runtime mask, so it is related infrastructure rather than a baseline. Sparse matrix multiplication (DTC-SpMM [11], MaxK-GNN [29], Sputnik [13]) reorganizes a sparse matrix operand so the dense datapath can consume it; its sparsity lives in the matrix structure of a different operator, not in a spatial mask over a convolution grid. Structured weight sparsity, such as 2:4 tensor cores [25], exploits fixed patterns in the weights. Sparse-format compilers such as TACO [20] and SparseTIR [41] instead target tensor algebra over explicitly sparse representations. Neither directly targets a dynamic output mask over a dense convolution grid. DPACS [14] resolves similar dynamic spatial sparsity in CNN feature maps, but on a custom accelerator instead of a commodity GPU.

3 WISEConv Overview

Dense tiled convolution already consumes a stream of output coordinates: each coordinate determines the receptive-field reads and output write, while its reduction over channels and kernel offsets remains regular. WISEConv exploits this interface by replacing the dense coordinate sequence with a worklist of requested output positions, thereby introducing position-level selectivity into the tiled pipeline. Removing unrequested positions from the schedule raises the useful-compute ratio, but issuing fewer operations alone does not guarantee lower latency. To translate selectivity into latency reduction, the worklist must be inexpensive to construct and must be consumed with both a high useful-compute ratio and high throughput. WISEConv therefore co-designs a construction stage and a compute stage around these interdependent requirements, as Figure 2 summarizes.

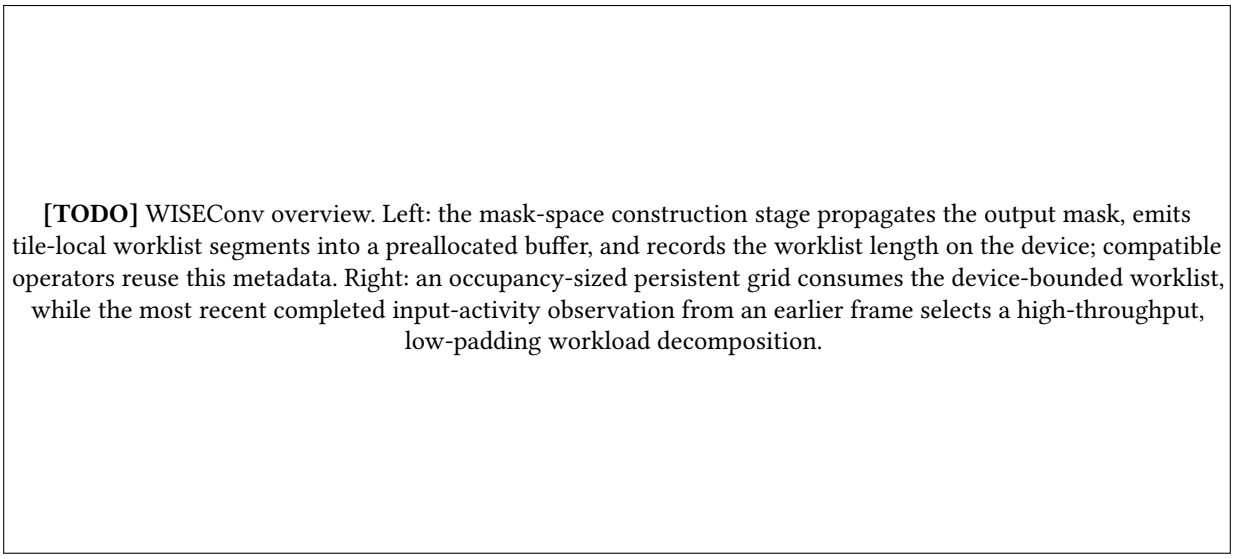


Figure 2. The WISEConv two-stage pipeline. Construction emits tile-locally ordered worklist segments using only mask and coordinate metadata, then reuses the metadata across compatible operators. Compute bounds persistent work on the device and adapts its decomposition from a completed, temporally lagged input-activity observation without host synchronization.

Construction stage. To prevent worklist construction from erasing the savings of selective compute, WISEConv confines construction to mask and coordinate metadata, never reading feature or weight tensors. In one mask-space pass, it fuses the output-mask propagation in Eq. 1 with tile-local compaction, producing M_{out} and a worklist $\mathcal{W}$; the requested coordinates from each spatial tile form a segment in local dense-traversal order, preserving spatial grouping without global ordering. WISEConv stores $\mathcal{W}$ in a buffer preallocated for $|\mathcal{G}| = H_{out}W_{out}$ coordinates, the static upper bound on $|\mathcal{W}|$. A device-resident counter records $n_{\mathcal{W}} = |\mathcal{W}|$, so construction needs neither runtime allocation nor a host-visible active count.

Even this metadata-only pass must scan the layer's mask grid. WISEConv therefore propagates and reuses the resulting mask and worklist metadata across compatible operators instead of reconstructing it, amortizing one construction across the compatible operator sequence. The fused pass lowers the cost of each construction, while cross-operator reuse reduces how often it occurs; together, they address the Worklist Construction Cost challenge.

Compute stage. The worklist length varies across layers and inputs and remains device-resident; synchronizing its current value to the host would stall the per-frame path, yet the workload decomposition must adapt because its padded work and exposed parallelism govern useful-compute ratio and GPU throughput. To execute without a host-visible bound, WISEConv launches only enough persistent thread blocks to fill the GPU's concurrent block residency and lets them consume $\mathcal{W}$ through a grid-stride loop bounded by device-resident $n_{\mathcal{W}}$, so the launch footprint tracks resident capacity rather than the full dense output grid.

To choose a decomposition without querying the exact $n_{\mathcal{W}}$, WISEConv builds from representative temporal traces a table that maps input-activity levels to configurations that sustain throughput while limiting padding, then indexes it with the most recent completed observation from one or more frames earlier; temporal correlation keeps this lagged signal informative, while completed-only polling prevents selection from blocking. The device count thus bounds the current work, while the data-driven signal chooses its decomposition; together, they sustain useful-compute ratio and GPU throughput as the worklist length varies without synchronizing the per-frame path.

Neither stage suffices alone: excessive construction cost, a low useful-compute ratio, or low consumption throughput can each erase the benefit of issuing less convolution work. By controlling all three jointly, WISEConv converts upstream spatial sparsity into end-to-end latency reduction.

4 WISEConv Design

The design follows the worklist from its construction and propagation across compatible operators (§4.1) to its consumption by the convolution pipeline (§4.2).

4.1 Worklist Construction and Reuse

4.1.1 Tile-Local Worklist Construction. Constructing a tight schedule already requires a scan of the output grid, but the resulting coordinates must also retain enough spatial structure for efficient consumption. Globally stable compaction preserves the complete dense traversal order but

requires a prefix scan or another pass to obtain device-wide offsets; per-coordinate appends avoid that coordination but scatter neighboring positions. Materializing active receptive fields, as gather-scatter does, regularizes the compute input at the cost of feature traffic proportional to the active count and $k^2 C_{in}$. The useful middle ground is therefore a metadata-only schedule with local rather than global ordering.

WISEConv realizes this middle ground with contiguous, tile-local worklist segments. Because the number of active positions is not known before the pass, it preallocates a coordinate buffer with capacity $|\mathcal{G}| = H_{out} W_{out}$, the static upper bound on $|\mathcal{W}|$. Algorithm 1 details the resulting single pass. Line 1 initializes the device-resident length counter. Lines 2–7 partition the output grid into $B_\tau \times B_\tau$ construction tiles, derive the corresponding entries of M_{out} using Eq. 1, and compact each tile's active coordinates into a local sequence $\mathcal{P}_\tau$ in dense-traversal order. When a learned gate supplies M_{out} directly [21, 35], Lines 4–5 are omitted and Line 7 compacts the supplied mask. Lines 8–11 count the compacted coordinates, atomically reserve $[\text{base}, \text{base} + n_\tau)$ from the counter, and copy $\mathcal{P}_\tau$ into that interval. After all tiles complete, the reserved intervals collectively store $\mathcal{W}$, and the counter records $n_{\mathcal{W}}$. Because the layer shape fixes the buffer capacity and launch geometry while only the device observes the data-dependent length, construction requires no runtime allocation or host-visible active count.

To relate the worklist to the useful-compute ratio of Eq. 3, we insert its length $n_{\mathcal{W}}$ between the required-position count and the issued-slot count. For $n_{\mathcal{W}} > 0$,

$$\eta = \underbrace{\frac{\|M_{out}\|_0}{n_{\mathcal{W}}}}_{\eta_{\text{construct}}} \cdot \underbrace{\frac{n_{\mathcal{W}}}{|C|}}_{\eta_{\text{compute}}} . \tag{4}$$

Here, $\eta_{\text{construct}}$ compares required positions with worklist entries, while η_{compute} compares worklist entries with issued slots. We analyze the former here and the latter after defining the compute organization in §4.2. The factorization decomposes η only; all stage costs remain reflected in throughput T.

Because construction tiles partition the output grid and use disjoint reservations, each active coordinate appears exactly once, so $n_{\mathcal{W}} = \|M_{out}\|_0$. Fresh construction therefore has $\eta_{\text{construct}} = 1$ for a nonempty worklist. Coverage constrains membership rather than order, so tile reservation order does not affect correctness; traversal order within each interval is retained for locality.

WISEConv thus confines its ordering guarantee to each contiguous segment: coordinates retain the tile's dense traversal, while independently reserved segments provide no cross-boundary locality guarantee. As shown in §6.4, this tile-local order achieves compute-stage throughput comparable to global ordering, making the additional ordering pass an unfavorable end-to-end trade-off.

Algorithm 1 Tile-Local Worklist Construction

Require: Input mask M_{in}, output grid $\mathcal{G}$, tile size B_τ
Ensure: Mask M_{out}, worklist $\mathcal{W}$, length $n_{\mathcal{W}} = |\mathcal{W}|$

1: $n_{\mathcal{W}} \leftarrow 0$
2: **for** each $B_\tau \times B_\tau$ spatial tile τ **in parallel do**
3: **for** each position $p \in \tau$ **in parallel do**
4: $\text{active}(p) \leftarrow \exists r \in \mathcal{N}(p) : M_{in}[r] = 1$
5: $M_{out}[p] \leftarrow \text{active}(p)$
6: **end for**
7: $\mathcal{P}_\tau \leftarrow \textsc{Compact}(\tau, M_{out})$
8: $n_\tau \leftarrow |\mathcal{P}_\tau|$
9: $\text{base} \leftarrow \text{AtomicAdd}(n_{\mathcal{W}}, n_\tau)$
10: **for** each $0 \le j < n_\tau$ **in parallel do**
11: $\mathcal{W}[\text{base} + j] \leftarrow \mathcal{P}_\tau[j]$
12: **end for**
13: **end for**

4.1.2 Cross-Operator Worklist Reuse.

Each construction pass examines all $H_{out} W_{out}$ positions and, when deriving M_{out} from M_{in}, performs $O(k^2)$ mask work per position. This channel-independent grid scan contrasts with required convolution work, which scales as $O(\alpha H_{out} W_{out} k^2 C_{in} C_{out})$. Repeating the scan across same-grid layers that introduce no new required positions is therefore redundant. WISEConv instead amortizes construction over each compatible layer sequence by reusing one worklist across its operators.

For layer l, let $\mathcal{G}^l$ denote its output grid, M^l_{out} its output mask, $\mathcal{W}^l$ its worklist, and $n^l_{\mathcal{W}}$ its device-resident length. WISEConv propagates the mask and worklist metadata alongside the feature tensor. From layer semantics, WISEConv statically infers that consecutive layers are reuse-compatible when $\mathcal{G}^{l+1} = \mathcal{G}^l$ and

$$\{p \in \mathcal{G}^{l+1} \mid M^{l+1}_{out}[p] = 1\} \subseteq \{p \in \mathcal{G}^l \mid M^l_{out}[p] = 1\}. \tag{5}$$

Because $\mathcal{W}^l$ covers M^l_{out}, Eq. 5 guarantees coverage for M^{l+1}_{out}. WISEConv therefore forwards $\mathcal{W}^{l+1} = \mathcal{W}^l$ and $n^{l+1}_{\mathcal{W}} = n^l_{\mathcal{W}}$ without reconstruction and applies the condition recursively at each subsequent layer.

Equality in Eq. 5 covers same-grid 1×1 convolutions, mask-preserving activations, and inference-time normalization, leaving $\eta_{\text{construct}}$ unchanged. Strict inclusion can result from mask-narrowing operators in the upstream policy, such as the activation-fused update truncation that DeltaCNN uses to increase downstream sparsity [28]. The inherited worklist remains a valid superset, trading a lower $\eta_{\text{construct}}$ for one fewer construction pass without violating coverage.

A change in the output grid violates the geometry constraint, while receptive-field expansion may violate Eq. 5 by introducing positions outside the upstream mask; either case invokes the construction pass of §4.1.1 to emit a new tight worklist. A lower $\eta_{\text{construct}}$ alone does not trigger reconstruction because tightening a valid superset would pay another grid scan. Common receptive-field-expanding layers such as

3×3 convolutions still create rebuild points throughout the network, leaving device-resident worklist lengths variable across layers and frames. Compute must therefore handle a dynamic problem size.

4.2 Variable-Length Worklist Execution

Construction and reuse fix $\eta_{\text{construct}}$; the implicit-GEMM decomposition controls η_{compute} and throughput T. For a standard ungrouped batch-1 convolution, dense implicit GEMM uses $M = H_{out}W_{out}$ output-position rows, $N = C_{out}$ output-channel columns, and reduction depth $K = k^2C_{in}$, with $B_M \times B_N$ output tiles and B_K-wide reduction steps. WISEConv preserves the N and K dimensions and this tiled pipeline, replacing only the M-dimension coordinate source with $\mathcal{W}$. Compute therefore groups $\mathcal{W}$ into B_M-coordinate tiles, padding only the final tile. For $n_{\mathcal{W}} > 0$, this issues $|C| = B_M\lceil n_{\mathcal{W}}/B_M\rceil$ position slots, giving

$$\eta_{\text{compute}} = \frac{n_{\mathcal{W}}}{B_M\,\lceil n_{\mathcal{W}}/B_M\rceil}. \tag{6}$$

The final tile contains at most $B_M - 1$ padding slots, making this overhead most pronounced for a short worklist. The same B_M also controls data reuse, block efficiency, and exposed parallelism, so maximizing η_{compute} need not maximize throughput T.

Because dense (M, N, K) follow from the layer shape, the backend can select (B_M, B_N, B_K) before execution. In the worklist path, however, the M-dimension extent is $n_{\mathcal{W}}$, which varies across invocations and remains unknown to the host. This leaves two problems: executing the current worklist without exposing its length to the host, and selecting a decomposition suited to that length. §4.2.1 addresses the former with persistent execution, while §4.2.2 addresses the latter through trace-calibrated selection.

4.2.1 Persistent Worklist-Driven Convolution. Dense row m obtains its coordinate p_m from the output-grid traversal; WISEConv instead sets $p_m = \mathcal{W}[m]$. Because p_m determines the receptive-field and output addresses, only the input gathers and output stores change. Feature and weight layouts and the regular B_K-tiled reduction remain unchanged, and no receptive-field patches are materialized. Contiguous ranges of $\mathcal{W}$ carry construction's tile-local grouping into the gathers.

Because the host chooses a conventional launch's thread-block count, using device-resident $n_{\mathcal{W}}$ would require a synchronizing transfer; launching for the output-grid upper bound instead submits blocks in proportion to dense work. WISEConv derives each configuration's persistent-grid capacity g_{cap} from the GPU's concurrent block residency and launches exactly that many blocks, independently of $n_{\mathcal{W}}$.

Algorithm 2 assigns work entirely on the device. Line 1 derives the number of position-channel work items from $n_{\mathcal{W}}$; Lines 2–3 distribute them across g_{cap} persistent blocks in a grid-stride loop. For each item, Lines 4–8 recover its (b_m, b_n) indices, load and pad the corresponding B_M-coordinate slice, select its B_N output channels, and initialize a fixed $B_M \times B_N$ accumulator. Lines 9–16 perform the regular implicit-GEMM reduction: each kernel offset and B_K-wide channel slice forms $\mathbf{X}$ and $\mathbf{W}$ directly from the dense tensors (Lines 12–13) and accumulates $\mathbf{XW}$ (Line 14). Line 17 writes the nonpadding rows to their worklist coordinates, after which Line 18 advances the block by g_{cap}. For clarity, the pseudocode omits the boundary checks required when C_{in} or C_{out} is not tile-aligned. Thus, $n_{\mathcal{W}}$ controls only the device-side work bound and final position padding; the multiply shape and tensor layouts remain fixed, leaving only the choice of an efficient decomposition for this bound.

Algorithm 2 Persistent Worklist-Driven Implicit GEMM

Require: Device-resident $(\mathcal{W}, n_{\mathcal{W}})$, persistent-grid capacity g_{cap}, tensors $(\mathbf{x}, \mathbf{w})$, and tile shape (B_M, B_N, B_K)
Ensure: Output $\mathbf{y}$ (updated at worklist coordinates)

1: $n_{\text{work}} \leftarrow \lceil n_{\mathcal{W}}/B_M\rceil\lceil C_{out}/B_N\rceil$
2: **for** $t_i = 0, \ldots, g_{\text{cap}} - 1$ **in parallel do**
3: **while** $t_i < n_{\text{work}}$ **do**
4: $(b_m, b_n) \leftarrow \text{Unflatten}(t_i)$
5: $\mathcal{P} \leftarrow \mathcal{W}[b_mB_M : \min((b_m + 1)B_M, n_{\mathcal{W}})]$
6: Pad $\mathcal{P}$ to B_M entries with $\perp$
7: $c_1 \leftarrow b_nB_N, \quad c_2 \leftarrow (b_n + 1)B_N$
8: $\mathbf{Y} \leftarrow \mathbf{0}_{B_M \times B_N}$
9: **for** $j = 1, \ldots, k^2$ **do**
10: **for** $r = 0, \ldots, \lceil C_{in}/B_K\rceil - 1$ **do**
11: $r_1 \leftarrow rB_K, \quad r_2 \leftarrow (r + 1)B_K$
12: $\mathbf{X} \leftarrow [\,\mathbf{x}[\mathcal{N}_j(p), r_1 : r_2]\,]_{p\in\mathcal{P}}$, zero-padding $\perp$
13: $\mathbf{W} \leftarrow \mathbf{w}_j[r_1 : r_2, c_1 : c_2]$
14: $\mathbf{Y} \leftarrow \mathbf{Y} + \mathbf{X} \cdot \mathbf{W}$
15: **end for**
16: **end for**
17: $\mathbf{y}[\mathcal{P}, c_1 : c_2] \leftarrow \mathbf{Y}$, skipping $\perp$
18: $t_i \leftarrow t_i + g_{\text{cap}}$
19: **end while**
20: **end for**

4.2.2 Trace-Calibrated Adaptive Decomposition. Persistent scheduling resolves the execution bound, but not the decomposition choice: for a fixed layer, the fastest configuration changes with $n_{\mathcal{W}}$. A configuration $q = (B_M, B_N, B_K, S_K)$ includes a split-K factor S_K, the number of disjoint partitions of the $K = k^2C_{in}$ reduction for each position-channel tile. Algorithm 2 presents the $S_K = 1$ path, which writes each completed reduction directly. For $S_K > 1$, standard split-K first zeros $\mathbf{y}$ at the scheduled coordinates, then computes the partitions independently and atomically accumulates their partial sums into $\mathbf{y}$. For layer l, the values supported on the target GPU define the configuration domain

$$\mathcal{Q}_l = \mathcal{B}_M^l \times \mathcal{B}_N^l \times \mathcal{B}_K^l \times \mathcal{S}_K^l,$$

where each factor contains the supported values of its corresponding parameter.

The length-sensitive effects are clearest in B_M and S_K. B_M trades final-tile padding and exposed position parallelism against within-tile reuse. Increasing S_K exposes S_K times as many work items when a short worklist would otherwise underfill the GPU, at the cost of coordinate rereads, output zeroing, and atomic accumulation. Because all split-K partitions cover the same coordinates, split-K affects throughput T but leaves C and η_{compute} unchanged.

WISEConv builds a level-to-configuration table $\mathcal{L}$, with one row $\mathcal{L}_l$ per layer, after quantizing α into D levels as $d = \max\{1, \lceil D\alpha \rceil\}$. Calibration deliberately represents each level with a fresh, tight worklist rather than reproducing every reuse state. For each layer l and level d, WISEConv constructs a synthetic worklist containing $\lceil d|\mathcal{G}^l|/D \rceil$ active coordinates with construction's tile-local structure, then measures $\widehat{t}_l(q; d)$, the latency of every $q \in Q_l$ at that level. It records

$$q^*_{l,d} = \arg\min_{q \in Q_l} \widehat{t}_l(q;\, d). \tag{7}$$

Because all worklist candidates execute the same synthetic worklist, Eq. 7 selects the minimum-latency decomposition, equivalently the one with the highest effective throughput at that level. WISEConv stores it as $\mathcal{L}_l[d] = q^*_{l,d}$. When full-grid execution is semantics-compatible, calibration also compares a dense candidate as a special case and stores it in the same table if it is faster.

Lookup is cheap once d is known, but obtaining every layer's current level directly would require transferring its device-resident activity count to the host. Waiting would synchronize each layer. Making these transfers asynchronous and using the latest completed values removes the waits, but the runtime must still enqueue and track one scalar transfer per layer per frame, so observation overhead scales with the layer count. WISEConv deliberately rejects both alternatives: it enqueues only one asynchronous transfer of network-input activity per frame, then uses the latest completed, necessarily lagged observation to predict every layer's level. For event and temporal workloads, this single observation remains predictive: layer masks derive from a shared network-input mask through receptive-field propagation, and successive input masks correlate in time.

To fit this predictor, WISEConv profiles a set $\mathcal{F}$ of representative trace frames for the set $\mathcal{I}$ of layers that use it, recording $\{\alpha_0^f\}_{f \in \mathcal{F}}$ and $\{\alpha_l^f\}_{(f,l) \in \mathcal{F} \times \mathcal{I}}$. It partitions the observed input-activity range $[\alpha_{\text{lo}}, \alpha_{\text{hi}}]$ into B equal-width buckets, with $\alpha_{\text{lo}} = \min_{f \in \mathcal{F}} \alpha_0^f$ and $\alpha_{\text{hi}} = \max_{f \in \mathcal{F}} \alpha_0^f$. Frame f receives index $b^f \in \{1, \ldots, B\}$:

$$b^f = \max\left\{1, \left\lceil \frac{B(\alpha_0^f - \alpha_{\text{lo}})}{\alpha_{\text{hi}} - \alpha_{\text{lo}}} \right\rceil\right\}. \tag{8}$$

Equal-width buckets assign fixed activity ranges independent of sample frequency and therefore preserve resolution in rare regimes.

Given these frame groups, WISEConv fits a bucket-to-level prediction table $\mathcal{A}$ with layer-specific rows $\mathcal{A}_l$, since grid resolution, receptive-field size, and propagation depth transform the same input activity differently at each layer. For every bucket b, WISEConv computes the median layer activity $\widetilde{\alpha}_{l,b} = \text{median}_{f \in \mathcal{F}:\, b^f = b}\, \alpha_l^f$ over the frames assigned to b, then quantizes it into the predicted level for layer l in bucket b:

$$\hat{d}_{l,b} = \max\left\{1, \left\lceil D\widetilde{\alpha}_{l,b} \right\rceil\right\}. \tag{9}$$

The median minimizes absolute deviation within the bucket and is robust to outliers. WISEConv stores $\mathcal{A}_l[b] = \hat{d}_{l,b}$ and fills empty buckets from their nearest populated neighbors.

At runtime, WISEConv composes $\mathcal{A}_l$ and $\mathcal{L}_l$ to translate the latest completed network-input activity observation into configurations for all layers. At frame f, if this observation comes from $f' < f$, WISEConv clamps $\alpha_0^{f'}$ to $[\alpha_{\text{lo}}, \alpha_{\text{hi}}]$ and applies Eq. 8 to obtain $\hat{b}^f$, avoiding extrapolation beyond trace coverage; if none has completed, it retains $\hat{b}^{f-1}$. It then selects

$$q_l^f = \mathcal{L}_l\left[\mathcal{A}_l[\hat{b}^f]\right]. \tag{10}$$

The observation may lag the current input, and reuse may make the runtime worklist longer than the predicted activity level implies. Both effects influence only the configuration: persistent compute still reads the current $n^l_{\mathcal{W}}$ on the device as the exact execution bound. Selection error can therefore affect performance but cannot omit scheduled coordinates.

Algorithm 3 connects the offline tables to nonblocking selection over the inference sequence $\mathcal{R}$. Lines 1–6 build $\mathcal{L}$ and $\mathcal{A}$ from the calibration and trace data; Lines 7–9 initialize the asynchronous queue and initial bucket. On each frame, Lines 10–17 retain the previous bucket if no observation has completed; otherwise, they clamp and bucket the latest value and compose $\mathcal{A}_l$ with $\mathcal{L}_l$ to select all layer configurations. Lines 18–21 derive and enqueue the current input active ratio asynchronously, and Line 22 executes the frame with the selected configurations. A value becomes visible only after its device-to-host transfer completes. The bounded queue returns the newest visible value and discards older ones; when no slot is free, enqueue drops the new observation. Neither operation waits, so selection never stalls construction or compute.

Learned-gating models lack the shared input mask required by this predictor: each gated block derives its own mask from intermediate features. Observing their activity levels asynchronously would require one transfer per gated layer per frame, so WISEConv instead assigns each layer a fixed level calibrated from its gate statistics. The trace-driven mode thus uses one nonblocking input observation

Algorithm 3 Trace-Calibrated Nonblocking Configuration Selection

Require: Calibration frames $\mathcal{F}$, inference sequence $\mathcal{R}$, layers $\mathcal{I}$, D, and B
Ensure: Inference results for $\mathcal{R}$

Offline:
$\mathcal{L} \leftarrow$ level-to-configuration table from Eq. 7
$\{\alpha_0^f\}_f, \{\alpha_l^f\}_{f,l} \leftarrow$ input and layer active-ratio traces
$(\alpha_{\text{lo}}, \alpha_{\text{hi}}) \leftarrow (\min_{f\in\mathcal{F}} \alpha_0^f, \max_{f\in\mathcal{F}} \alpha_0^f)$
$\{b^f\}_f \leftarrow$ input buckets of $\{\alpha_0^f\}_f$ by Eq. 8
$\mathcal{A} \leftarrow$ bucket-to-level table from Eq. 9
Online:
Queue $\leftarrow$ empty asynchronous queue
$\hat{b}^0 \leftarrow$ calibrated initial input bucket
for inference frame $f = 1, \ldots, |\mathcal{R}|$ **do**
  $\hat{b}^f \leftarrow \hat{b}^{f-1}$
  if *Queue* is not empty **then**
    $\alpha_0^{f'} \leftarrow$ dequeueLatest(*Queue*) ▷ $f' < f$
    $\alpha_0^{f'} \leftarrow \text{clamp}(\alpha_0^{f'}, \alpha_{\text{lo}}, \alpha_{\text{hi}})$
    $\hat{b}^f \leftarrow$ input bucket of $\alpha_0^{f'}$ by Eq. 8
  end if
  $\{q_l^f\}_l \leftarrow \{\mathcal{L}_l[\mathcal{A}_l[\hat{b}^f]]\}_l$
  async do
    $\alpha_0^f \leftarrow$ current input-mask active ratio (Eq. 3)
    Enqueue α_0^f into *Queue*
  end async
  Run frame f with $\{q_l^f\}_l$ via Algorithms 1 and 2
end for

per frame, while the fixed-level mode uses none; neither adds per-layer device-to-host transfers to frame execution.

5 Implementation

We implement WISEConv as a PyTorch extension with a Python frontend and a CUDA backend written in C++. A model-conversion pass replaces supported modules, preserves their parameters and weights, and statically marks worklist-reuse edges and rebuild points. The backend operates on FP32 tensors in NHWC layout and implements the standard, grouped, and transposed convolutions and mask-aware auxiliary operators required by our workloads. Each converted output carries a lightweight record that references its device-resident M_{out}, $\mathcal{W}$, and $n_{\mathcal{W}}$ buffers. All buffers are allocated during module initialization, while mask selection remains with the upstream application.

The construction kernel maps one CUDA block to each construction tile in Algorithm 1. Warp ballots compute lane-local ranks, a short shared-memory prefix combines the warp counts, and one `atomicAdd` reserves the tile's contiguous worklist segment. Worklists use 32-bit linear coordinates and are provisioned for $|\mathcal{G}|$ entries; the runtime resets their device counters on the operator's CUDA stream. Learned-gating layers compact the supplied M_{out} directly, and compatible downstream operators implement reuse by aliasing the existing metadata buffers.

The compute backend instantiates a C++ template family over (B_M, B_N, B_K, S_K) and derives each variant's g_{cap} from CUDA occupancy information and the GPU's SM count. The online selector in Algorithm 3 uses a bounded queue of pinned host scalars, CUDA events, and a dedicated copy stream. It transfers one network-input count per frame, updates the selected variant only from the newest completed observation, and drops an observation when no queue slot is free. Each invocation launches the selected variant's fixed persistent grid, while the kernel reads the current $n_{\mathcal{W}}$ to bound its grid-stride loop. Every backend variant agrees numerically with cuDNN at each active output position.

6 Evaluation

6.1 Experimental Setup

Platforms. We evaluate six GPU platforms spanning four discrete GPUs and two embedded SoCs. The discrete platforms are RTX 4070 Laptop, RTX 3080, *TBD Discrete GPU A*, and *TBD Discrete GPU B*. The four discrete GPUs cover a range of compute capabilities. Jetson Xavier NX and Jetson AGX Orin extend the evaluation to embedded SoCs with limited compute and power budgets, where we also measure board energy.

Models, tasks, and mask sources. Our workloads span three vision tasks.

- **Event-based optical flow.** FireFlowNet [26] is a lightweight event-based optical-flow model that delivers high inference rates while retaining competitive accuracy. We evaluate it on MVSEC [43], which contains event-camera sequences and corresponding ground-truth flow from indoor quadrotor flight and outdoor driving. Following Ev-Conv [7], consecutive 50-ms windows advance by 1 ms; events entering or leaving the window form the sparse increment from which we derive layer masks.
- **Detection.** We evaluate YOLOv8n and YOLOv8m [18] on MOT16 [24], a collection of crowded pedestrian videos for multi-object tracking. Their two model sizes expose how scale affects sparse-execution efficiency. For both models, DeltaCNN's temporal policy [28] thresholds activation differences between successive frames and propagates the resulting masks across layers.
- **Human pose estimation.** We use the DynConv pose model [35] on MPII [1], which depicts people performing diverse activities and annotates their 2D body joints. DynConv's learned spatial gates produce an input-dependent mask for each gated block.

Table 1 summarizes the workloads and evaluation scales. For every workload, calibration uses the training split, while all reported results use a disjoint test split. We use the authors' released weights for all four models [18, 26, 35]. Within each workload, we fix the inputs, weights, and upstream mask policy; every sparse path receives the same resulting layer masks, isolating the masked-convolution backend.

Table 1. Evaluation workloads.

	FireFlowNet [26]	**YOLOv8n/m** [18]	**DynConv** [35]
Task	Optical flow	Detection	Human pose
Mask source	Ev-Conv [7]	DeltaCNN [28]	DynConv [35]
Dataset	MVSEC [43]	MOT16 [24]	MPII [1]
Input size	256×256	640×640	256×256
Parameters	0.056M	3.16M / 25.9M	6.89M
Eval. samples	196.8K	2,575	2,958

Baselines. We compare WISEConv against three FP32 execution paths established by prior work:

- **Dense** executes the complete model with the native PyTorch/cuDNN backend [5].
- **Tile skipping** adapts DeltaCNN's CUDA implementation [28], retaining the dense tiled pipeline and skipping a tile only when all its outputs are inactive.
- **Gather-scatter** uses DynConv's original CUDA backend for pose [35]. FireFlowNet and YOLO use an adaptation of MMCV's masked-convolution operator [4].

Measurements. We measure per-frame full-model GPU latency over the complete evaluation data above with CUDA events; every sequence or split is repeated three times. Timing covers mask propagation, worklist construction, all network operators, and output update; it excludes input transfer, offline calibration and autotuning, and one-time state initialization. On the Jetson platforms, we integrate board-level `VDD_IN` over each timed round to obtain its mean power, multiply that value by the round's cumulative CUDA-event time, and aggregate across rounds to obtain per-frame board energy. We follow the original workload protocols: AEE for optical flow [7], COCO-style average precision (AP) for detection, averaged over IoU thresholds from 0.50 to 0.95 [28], and PCKh for human pose [35]. Because MOT16 releases ground-truth detections only for its training split, AP on the test sequences uses fixed pseudo-labels from a dense YOLO26x teacher [19].

6.2 End-to-End Performance

Latency. WISEConv is the fastest path in all 16 measured workload-platform pairs (Figure 3). Relative to the fastest competing path for each pair, it reduces full-model latency by 34.2–46.5% on the discrete GPUs and 22.1–38.6% on the Jetson SoCs. These gains persist across all four workloads and both device classes despite a more than two-order-of-magnitude span in absolute latency, from 0.399 ms for FireFlowNet on RTX 3080 to 94.3 ms for YOLOv8m on Xavier NX.

Existing sparse paths realize the available headroom only in narrow regimes: tile skipping needs low activity to offset tile-level waste, while gather-scatter needs sufficiently heavy convolution to amortize its overhead. FireFlowNet and DynConv leave enough removable work for tile skipping to cut dense latency by up to 42.0% and 40.9%. The higher-activity DeltaCNN masks for YOLOv8n and YOLOv8m leave less headroom, and tile skipping fails to turn it into a material speedup: it is near parity with dense on AGX Orin and slower in the other six pairs. Gather-scatter becomes the fastest existing path only for YOLOv8m on AGX Orin, where the heavier convolution makes the arithmetic saved by exact position selection outweigh gather-scatter's extraction and writeback overhead. Despite these regime-dependent baseline gains, WISEConv remains fastest in every pair.

Across all workloads, WISEConv is substantially closer to the ideal reference than the fastest existing path. For FireFlowNet and the two YOLO models, WISEConv runs at 1.19–2.10× the reference, while the fastest existing path remains at 1.88–3.21×. As the active ratio falls, active convolution work shrinks while mask processing, worklist construction, and other fixed model work do not, magnifying these costs relative to the ideal reference. DynConv makes this effect most visible: it has the lowest active ratio and additionally runs learned mask-prediction kernels in every gated block. Even so, WISEConv runs at 2.34–3.66× the reference, while the fastest existing path remains at 3.80–6.43×.

Energy. WISEConv's latency gains also reduce per-frame board energy on both Jetson platforms. It is the lowest-energy path for every workload (Table 2), reducing energy by 17.0–39.8% relative to the lowest-energy competing path in each pair and by 28.5–72.6% relative to dense execution.

Accuracy. WISEConv's latency and energy gains preserve the accuracy of the selected sparsity policies. For FireFlowNet and DynConv, its AEE and PCKh match the other

Table 2. Per-frame board energy (J) on the Jetson platforms. Parentheses report each sparse path's change relative to Dense.

GPU	System	FireFlowNet	YOLOv8n	YOLOv8m	DynConv
AGX Orin	Dense	0.26	0.48	2.70	1.20
	Tile skip	0.15 (-43.8%)	0.44 (-9.1%)	2.33 (-13.7%)	0.67 (-44.7%)
	Gather	0.29 (+9.5%)	0.70 (+46.1%)	2.42 (-10.4%)	2.37 (+97.0%)
	WISEConv	**0.10** (-61.6%)	**0.32** (-33.9%)	**1.57** (-41.7%)	**0.48** (-60.4%)
Xavier NX	Dense	0.28	0.49	2.58	1.08
	Tile skip	0.13 (-54.5%)	0.42 (-13.8%)	2.27 (-12.2%)	0.54 (-50.4%)
	Gather	0.33 (+17.0%)	0.85 (+75.8%)	3.18 (+23.2%)	2.83 (+160.9%)
	WISEConv	**0.08** (-72.6%)	**0.35** (-28.5%)	**1.78** (-31.1%)	**0.34** (-68.5%)

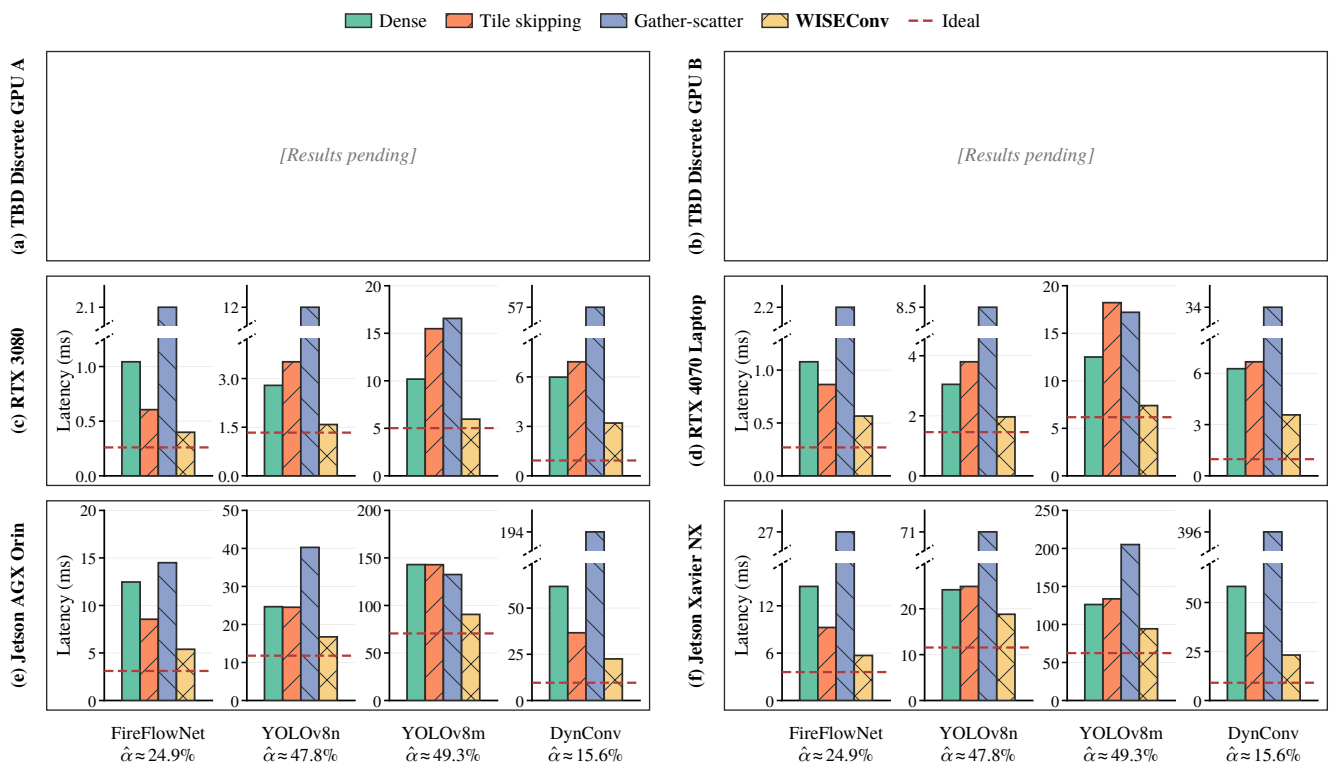


Figure 3. Full-model mean per-frame latency across GPU platforms. Each workload uses an independent y-axis. Labels report $\hat{\alpha}$, the model-level ratio of required to dense convolution FLOPs, accumulated over frames and layers and averaged across evaluation sequences. Red dashed lines mark ideal proportional scaling, computed by applying each sequence's ratio to its dense latency before aggregation.

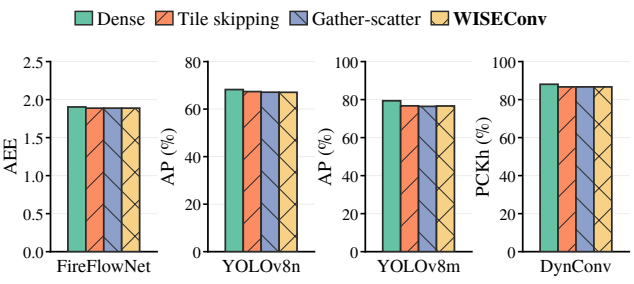


Figure 4. Task accuracy on RTX 3080 under the evaluated upstream sparsity policies, measured by AEE (↓), AP (↑), and PCKh (↑).

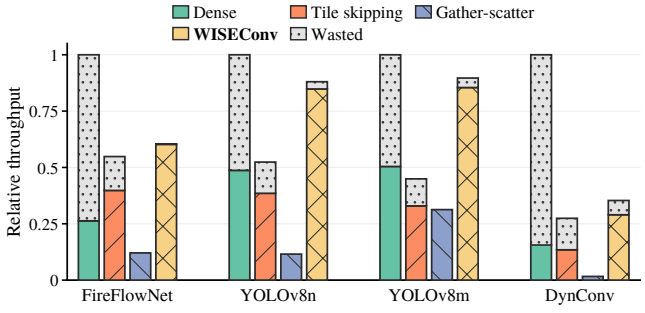


Figure 5. Full-model effective throughput ηT and wasted throughput $(1-\eta)T$ on RTX 3080, normalized per frame to dense raw throughput and averaged across layers and frames.

sparse paths at the reported precision. In the YOLOv8 models, small floating-point differences accumulate across frames under temporal execution; even so, WISEConv's AP differs from the other sparse paths by at most 0.5% in relative terms (Figure 4). Compared with dense execution, the upstream sparsity policies lower AP and PCKh by only 1.6–3.5%, while lowering FireFlowNet's AEE by 0.8%.

6.3 Microbenchmarks

Effective throughput. Under the same execution settings as the end-to-end evaluation, we estimate full-model throughput T and effective throughput ηT on RTX 3080 (Figure 5). Table 3 reports WISEConv's η decomposition and worklist-length distribution. WISEConv sustains 35.3–89.7%

Table 3. WISEConv η decomposition and worklist-length statistics on RTX 3080. The η values are averaged over layers and frames; $n_{\mathcal{W}}$ percentiles are taken over all layer–frame samples.

Workload	$\eta_{\text{construct}}$	η_{compute}	η	$n_{\mathcal{W}}$		
				P10	P50	P90
FireFlowNet	99.9%	99.1%	98.9%	1,303	15,083	33,880
YOLOv8n	98.5%	97.7%	96.2%	77	937	5,937
YOLOv8m	98.7%	96.5%	95.2%	75	840	6,110
DynConv	100.0%	81.7%	81.7%	3	43	389

of dense throughput, the highest among sparse paths in every workload at 1.10–1.99× that of the next-highest sparse path. DeltaCNN's lower raw throughput reflects the impact of routing very sparse tiles through its fused selective path [28]. The two YOLO workloads lie at the upper end because their higher active ratios expose more parallelism and amortize fixed costs.

WISEConv's useful-compute ratio is 98.9%, 96.2%, and 95.2% for FireFlowNet, YOLOv8n, and YOLOv8m. DynConv is lower at 81.7%; its low active ratio and small spatial grids yield exceptionally short worklists, with P10, P50, and P90 lengths of 3, 43, and 389. Padding in the issued position sequence C is therefore difficult to avoid. DynConv's layer-specific gates also provide no shared activity signal across layers, and the evaluated MPII [1] samples are not temporally related and thus cannot support lagged prediction; yet its useful-compute ratio remains well above tile skipping's 48.9%. Combining throughput and useful-compute ratio, WISEConv achieves the highest effective throughput in every workload, reaching 1.51–1.86× that of the strongest competing path.

Table 3 localizes the remaining wasted work: despite cross-operator reuse, $\eta_{\text{construct}}$ remains at 98.5–100.0%. The selected decompositions give η_{compute} values of 99.1%, 97.7%, and 96.5% for FireFlowNet, YOLOv8n, and YOLOv8m; because DynConv has $\eta_{\text{construct}} = 100.0\%$, its lower η comes entirely from compute-stage padding of its short worklists.

Latency cost decomposition. We decompose full-model YOLOv8n latency on RTX 3080 by execution stage (Table 4), where frames from MOT16 [24] are classified into bins of low or high active ratios. WISEConv's construction time remains at 0.105–0.106 ms while convolution grows from 1.09 to 1.29 ms across the two bins; construction's latency share consequently falls from 7.1% to 6.2%. Gather-scatter shows why exact selection alone is insufficient: it has the lowest convolution time at 0.73–0.83 ms, but extraction, setup, and writeback put 10.9–11.0 ms (92.2–93.1% of total latency) in *Other*. WISEConv instead spends 0.160–0.165 ms (9.8–10.9% of latency) on *Other*, with convolution dominant at 73.8–76.3%; direct worklist consumption avoids the gather-scatter path's data-movement bottleneck.

Table 4. Full-model YOLOv8n latency decomposition on RTX 3080 into worklist construction (Cons.), convolution (Conv.), elementwise operators (Elem.), and other work, with stage entries in ms and percentages of total latency in parentheses.

	System	Tot.	Cons.	Conv.	Elem.	Other
Low (40.7%)	Dense	2.79	—	2.54(91.2%)	0.14(5.0%)	0.10(3.7%)
	Tile skipping	3.42	—	3.09(90.3%)	0.16(4.6%)	0.17(5.1%)
	Gather	11.81	—	0.73(6.2%)	0.09(0.7%)	11.00(93.1%)
	WISEConv	1.47	0.10(7.1%)	1.09(73.8%)	0.12(8.2%)	0.16(10.9%)
High (56.7%)	Dense	2.79	—	2.54(91.2%)	0.14(5.0%)	0.10(3.7%)
	Tile skipping	3.59	—	3.25(90.4%)	0.17(4.7%)	0.18(4.9%)
	Gather	11.82	—	0.83(7.0%)	0.09(0.8%)	10.90(92.2%)
	WISEConv	1.69	0.11(6.2%)	1.29(76.3%)	0.13(7.7%)	0.17(9.8%)

Notes. Low and High median-split the evaluated MOT16 frames by active ratio; labels report bin means. Each stage time multiplies its per-frame NSYS fraction by the corresponding end-to-end CUDA-event latency.

6.4 Ablation Study and Sensitivity Analysis

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